

**23MAT117-Linear Algebra**  
**Lab Session Sheet - 9**  
**Eigenvalues, Eigenvectors and characteristic polynomial**

**Practice Questions**

1) Generate random integer square matrices  $A$  of order 4,  $B$  of order 10, and  $C$  of order 15. Find their eigenvalues and eigenvectors.

a) Verify the following properties of eigenvalues for  $A$ ,  $B$  and  $C$ :

i. Trace of a matrix = sum of eigenvalues of the matrix.

```
A = randi([-10, 10], 4, 4);  
B = randi([-10, 10], 10, 10);  
C = randi([-10, 10], 15, 15);
```

Generate random square matrices  $A$ ,  $B$ , and  $C$

```
[eigenvectors, eigenvalues] = eig(A);  
[eigenvectors, eigenvalues] = eig(B);  
[eigenvectors, eigenvalues] = eig(C);
```

Find their eigenvalues and eigenvectors

```
eigA = eig(A);  
eigB = eig(B);  
eigC = eig(C);
```

Getting only the eigenvalues separately.

```
traceA = trace(A)  
sumEigA = sum(eigA)  
traceB = trace(B)  
sumEigB = sum(eigB)  
traceC = trace(C)  
sumEigC = sum(eigC)
```

Calculate trace and sum of eigenvalues

Check if trace equals sum of eigenvalues for all matrices.

```
if round(traceA) == round(sumEigA) && round(traceB) == round(sumEigB) && round(traceC) == round(sumEigC)  
    disp('Trace equals sum of eigenvalues for all matrices.');
```

```
else
```

```
    disp('Trace does not equal sum of eigenvalues for all matrices.');
```

```
end
```

**OUTPUT:**

traceA = 13

sumEigA = 13

traceB = 29

sumEigB = 29.0000

traceC = 13

sumEigC = 13.0000

Trace equals sum of eigenvalues for all matrices.

**ii. Det of a matrix = Product of eigenvalues of the matrix.**

```
D1 = det(A)
P1 = prod(eigA)
D2 = det(B)
P2 = prod(eigB)
D3 = det(C)
P3 = prod(eigC)
```

Calculate determinant as product of eigenvalues

Check if trace equals sum of eigenvalues for all matrices.

```
if round(D1) == round(P1) && round(D2) == round(P2) && round(D3) == round(P3)
    disp('Determinant equals product of eigenvalues for all matrices.');
```

else

```
    disp('Determinant does not equal product of eigenvalues for all matrices.');
```

end

**OUTPUT:**

D1 = -1884

P1 = -1.8840e+03

D2 = 1.5142e+11

P2 = 1.5142e+11

D3 = -8.6487e+15

P3 = -8.6487e+15 + 1.0000e+00i

Determinant does not equal product of eigenvalues for all matrices.

**iii. Complex eigenvalues of a matrix appear in pairs.**

Check if complex eigenvalues appear in conjugate pairs for A.

```
if isequal(sort(conj(eigenvalues_A(imag(eigenvalues_A)~=0))),sort(eigenvalues_A
(imag(eigenvalues_A)~=0)))
    disp('Complex eigenvalues of A appear in conjugate pairs.');
```

else

```
    disp('Complex eigenvalues of A do NOT appear in conjugate pairs.');
```

end

Check if complex eigenvalues appear in conjugate pairs for B.

```
if isequal(sort(conj(eigenvalues_B(imag(eigenvalues_B)~=0))),
sort(eigenvalues_B(imag(eigenvalues_B)~=0)))
    disp('Complex eigenvalues of B appear in conjugate pairs.');
```

else

```
disp('Complex eigenvalues of B do NOT appear in conjugate pairs.');
```

```
end
```

Check if complex eigenvalues appear in conjugate pairs for C.

```
if isequal(sort(conj(eigenvalues_C(imag(eigenvalues_C)~=0))),
sort(eigenvalues_C(imag(eigenvalues_C)~=0)))
    disp('Complex eigenvalues of C appear in conjugate pairs.');
```

```
else
    disp('Complex eigenvalues of C do NOT appear in conjugate pairs.');
```

```
end
```

### **OUTPUT:**

Complex eigenvalues of A appear in conjugate pairs.  
Complex eigenvalues of B appear in conjugate pairs.  
Complex eigenvalues of C appear in conjugate pairs.

### **OR (CHECK IF COMPLEX EIGENVALUES OF ALL MATRICES APPEAR IN PAIRS AT ONCE)**

```
eigenvalues_A = eig(A);
eigenvalues_B = eig(B);
eigenvalues_C = eig(C);

for M = {eigenvalues_A, eigenvalues_B, eigenvalues_C}
    if isequal(sort(conj(M{1}(imag(M{1})~=0))), sort(M{1}(imag(M{1})~=0)))
        disp('Complex eigenvalues appear in conjugate pairs.');
```

```
    else
        disp('Complex eigenvalues do NOT appear in conjugate pairs.');
```

```
    end
end
```

### **OUTPUT:**

Complex eigenvalues appear in conjugate pairs.  
Complex eigenvalues appear in conjugate pairs.  
Complex eigenvalues appear in conjugate pairs.

**b) Find eigenvalues of  $5A$ ,  $B/3$  and  $C^{-1}$ . Compare them with eigenvalues of  $A$ ,  $B$  and  $C$  respectively. What do you observe?**

```
A = randi([-10, 10], 4, 4);
B = randi([-10, 10], 10, 10);
C = randi([-10, 10], 15, 15);
```

Generate random square matrices A, B and C.

```
eig(A);
eig(B);
eig(C);
```

Generate eigen values of A, B and C.

```
P= 5*A;
P1= eig(P)
Q=(B/3);
Q1=eig(Q)
R=C^(-1);
R1=eig(R)
```

Find eigenvalues of 5A, B/3 and C<sup>-1</sup>.

### **OUTPUT:**

```
P1 = 4×1 complex
-13.6151 +45.4400i
-13.6151 -45.4400i
28.6151 +24.7872i
28.6151 -24.7872i
Q1 =
3.8899 + 5.2972i
3.8899 - 5.2972i
6.3231 + 0.0000i
-0.9774 + 5.3622i
-0.9774 - 5.3622i
-4.0567 + 2.4525i
-4.0567 - 2.4525i
-1.6456 + 0.0000i
0.9721 + 0.2450i
0.9721 - 0.2450i
```

```
R1 =
0.3993 + 0.0000i
0.0876 + 0.0697i
0.0876 - 0.0697i
-0.0679 + 0.0000i
-0.0429 + 0.0309i
-0.0429 - 0.0309i
-0.0145 + 0.0716i
-0.0145 - 0.0716i
0.0437 + 0.0288i
0.0437 - 0.0288i
0.0387+ 0.0000i
-0.0063 + 0.0516i
-0.0063 - 0.0516i
0.0187 + 0.0475i
0.0187 - 0.0475i
```

**OBSERVATION:** The eigen values are complex conjugate pairs.

**c) Find eigenvectors of 5A, B/3 and C<sup>-1</sup>. Compare them with eigenvectors of A, B and C respectively. What do you observe?**

```
[eigVecP, ~] = eig(P)
[eigVecQ, ~] = eig(Q)
[eigVecR, ~] = eig(R)
```

### **OUTPUT:**

```
eigVecP =
-0.4284 - 0.1605i   -0.4284 + 0.1605i   0.3152 + 0.4197i   0.3152 - 0.4197i
0.7018 + 0.0000i   0.7018 + 0.0000i   0.1500 + 0.0674i   0.1500 - 0.0674i
```

$0.0377 + 0.3104i$      $0.0377 - 0.3104i$      $0.5926 + 0.0000i$      $0.5926 + 0.0000i$   
 $0.0201 + 0.4473i$      $0.0201 - 0.4473i$      $0.3060 + 0.5026i$      $0.3060 - 0.5026i$

eigVecQ =

|                   |                   |                   |                   |                   |                   |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| -0.0781 + 0.0495i | -0.0781 - 0.0495i | 0.0275 + 0.0000i  | 0.4722 + 0.0000i  | 0.4722 + 0.0000i  | 0.2333 - 0.2202i  | 0.2333 + 0.2202i  | 0.0997 + 0.0000i  | -0.2844 - 0.1016i | -0.2844 + 0.1016i |
| -0.5623 + 0.0000i | -0.5623 + 0.0000i | -0.1515 + 0.0000i | 0.0199 + 0.0370i  | 0.0199 - 0.0370i  | 0.1840 - 0.0147i  | 0.1840 + 0.0147i  | 0.1139 + 0.0000i  | -0.2514 - 0.1155i | -0.2514 + 0.1155i |
| 0.0970 - 0.3857i  | 0.0970 + 0.3857i  | 0.3644 + 0.0000i  | -0.0190 + 0.3763i | -0.0190 - 0.3763i | 0.0228 + 0.0824i  | 0.0228 - 0.0824i  | -0.1305 + 0.0000i | 0.4184 - 0.1145i  | 0.4184 + 0.1145i  |
| 0.1449 + 0.0258i  | 0.1449 - 0.0258i  | 0.0317 + 0.0000i  | -0.2733 - 0.2334i | -0.2733 + 0.2334i | 0.4931 + 0.0000i  | 0.4931 + 0.0000i  | -0.2848 + 0.0000i | -0.1256 - 0.0839i | -0.1256 + 0.0839i |
| 0.1263 + 0.2868i  | 0.1263 - 0.2868i  | -0.4761 + 0.0000i | 0.0212 + 0.1141i  | 0.0212 - 0.1141i  | -0.2731 - 0.2837i | -0.2731 + 0.2837i | -0.0161 + 0.0000i | 0.6413 + 0.0000i  | 0.6413 + 0.0000i  |
| -0.0239 - 0.0795i | -0.0239 + 0.0795i | 0.4272 + 0.0000i  | 0.1212 + 0.4146i  | 0.1212 - 0.4146i  | 0.1139 - 0.1382i  | 0.1139 + 0.1382i  | -0.1834 + 0.0000i | -0.1465 + 0.1518i | -0.1465 - 0.1518i |
| 0.1090 - 0.4114i  | 0.1090 + 0.4114i  | -0.4897 + 0.0000i | -0.0543 + 0.2300i | -0.0543 - 0.2300i | -0.0019 - 0.1940i | -0.0019 + 0.1940i | -0.4947 + 0.0000i | -0.2835 - 0.0968i | -0.2835 + 0.0968i |
| -0.2696 + 0.0348i | -0.2696 - 0.0348i | 0.2481 + 0.0000i  | 0.2606 - 0.0158i  | 0.2606 + 0.0158i  | 0.0097 + 0.1654i  | 0.0097 - 0.1654i  | -0.7417 + 0.0000i | -0.0653 + 0.0024i | -0.0653 - 0.0024i |
| 0.2164 + 0.1170i  | 0.2164 - 0.1170i  | 0.0386 + 0.0000i  | -0.2351 + 0.0275i | -0.2351 - 0.0275i | -0.3485 - 0.0358i | -0.3485 + 0.0358i | -0.0073 + 0.0000i | -0.1626 - 0.0695i | -0.1626 + 0.0695i |
| -0.2545 - 0.0998i | -0.2545 + 0.0998i | 0.3612 + 0.0000i  | 0.3196 - 0.1482i  | 0.3196 + 0.1482i  | -0.2071 - 0.4412i | -0.2071 + 0.4412i | -0.2240 + 0.0000i | -0.1640 + 0.1154i | -0.1640 - 0.1154i |

**OBSERVATION:** The eigenvectors appear in complex conjugate pairs.

**d) From  $A$ , using  $S = A + A'$  generate a symmetric matrix. Find the eigenvalues and eigenvectors of  $S$ .**

```
A = randi([-10, 10], 4, 4);
S=A+A'
```

```
[eigenvectors, eigenvalues] = eig(S)
```

**OUTPUT:**

```
S = -8  -7  15  2
     -7  -6  15  4
     15  15  -6  7
     2   4   7  10
```

```
eigenvectors = 0.5596  0.6361 -0.4564 0.2719
                0.5234 -0.7448 -0.1721 0.3765
                -0.6419 -0.0433 -0.5040 0.5763
                0.0307  0.1969  0.7128 0.6725
```

```
eigenvalues = -31.6437    0    0    0
                0  -0.2065    0    0
                0    0  2.8037    0
                0    0    0  19.0466
```

**2) Generate a random integer square matrix A of order 3.**

**a) Find the eigenvalues, right eigenvectors, and left eigenvectors of this matrix.**

```
A = randi([-5,5],3);
[R,L]=eig(A)
eig(A)
```

**OUTPUT:**

```
R = 0.5113  0.0656  0.4617
     0.7467 -0.8628 -0.6006
     0.4256 -0.5013  0.6528
```

```
L = 4.2193    0    0
     0  2.7533    0
     0    0 -6.9726
```

```
eigA= 4.2193
```

2.7533  
-6.9726

**b) Find the characteristic polynomial for this matrix.**

```
A = randi([-5,5],3)  
syms x;  
eq1=charpoly(A, x)
```

**OUTPUT:**

```
A = 1   3   7  
     4   5   2  
     9   2   5
```

```
eq1 = x^3 - 11*x^2 - 44*x + 244
```

**c) Replace x in the characteristic polynomial by the matrix A and evaluate the value. What is the value obtained? Try this with other square matrices.**

```
Eq2=subs(eq1,A)
```

**OUTPUT:**

```
Eq2 = [ 190, 40, -260]  
       [-44, -126, 120]  
       [-314, 120, -126]
```

**3) Generate an integer matrix A of order 5×3. Find  $AA^T$  and  $A^TA$ . Evaluate the eigenvalues of  $AA^T$  and  $A^TA$ . Verify the result that “The non-zero eigenvalues of  $AA^T$  and  $A^TA$  are same”.**

```
A=randi([0,9],5,3);  
rank(A);  
A1=A*A'  
A2=A'*A
```

Compute Eigenvalues for both the matrices.

```
E_A1=eig(A1)  
E_A2=eig(A2)
```

Remove near-zero eigenvalues for comparison.

```
E_A1_non_zero = E_A1(E_A1 > 1e-10)  
E_A2_non_zero = E_A2(E_A2 > 1e-10)
```

Compare non-zero eigenvalues.

```
if all(abs(E_A1_non_zero - E_A2_non_zero) < 1e-10)  
    disp('The non-zero eigenvalues of A*A'' and A''*A are the same.');
```

else

```
    disp('The non-zero eigenvalues of A*A'' and A''*A are NOT the same.');
```

end

**OUTPUT:**

```
A1 = 161 133 107 97 100
      133 125 89 83 118
      107 89 83 65 69
      97 83 65 59 67
      100 118 69 67 145
```

```
A2 = 180 116 158
      116 167 182
      158 182 226
```

```
E_A1 = -0.0000
        0.0000
        9.2180
        63.4386
        500.3435
```

```
E_A2 = 9.2180
        63.4386
        500.3435
```

```
E_A1_non_zero = 9.2180
                63.4386
                500.3435
```

```
E_A2_non_zero = 9.2180
                63.4386
                500.3435
```

The non-zero eigenvalues of  $A \cdot A'$  and  $A' \cdot A$  are the same.

**4) Generate a matrix with eigenvalues as 3,5,7,11 and eigenvectors as [1,1,2,2], [1,0,1,0], [2,1,2,0] and [0,1,1,0].**

```
Ld1 = 3;
Ld2 = 5;
Ld3 = 7;
Ld4 = 11;
```

Enter the given eigenvalues.

```
Ev1 = [1;1;2;2];
Ev2 = [1;0;1;0];
Ev3 = [2;1;2;0];
Ev4 = [0;1;1;0];
```

Enter the given eigenvectors.

```
V = [Ev1,Ev2,Ev3,Ev4];
D = diag([Ld1,Ld2,Ld3,Ld4]);
A = V*D*inv(V)
```

**OUTPUT:**

```
A = 9 4 -4 -1
     -4 7 4 -4
     -2 4 7 -5
     0 0 0 3
```

**5) Using MATLAB generate a  $5 \times 5$  matrix  $A$  of rank 2.**

**a) Obtain a symmetric matrix  $B = A + A'$  and find rank of  $B$ .**

```
A = randi([0 9], 5, 2)*randi([0 9], 2,5);
B=A+A'
rank(B)
```

**OUTPUT:**

```
B = 154 168 66 184 49
     168 198 68 218 83
        66 68 10 58 19
     184 218 58 226 95
        49 83 19 95 56
```

Rank = 4

**b) Find the eigenvalues and eigenvectors of  $B$  using MATLAB.**

```
[eigenvectors, eigenvalues] = eig(B)
```

**OUTPUT:**

```
eigenvectors = -0.4168    -0.4883    0.2161   -0.5560    0.4817
               -0.2962    0.4891   -0.5927    0.0515    0.5649
               0.7115   -0.3667   -0.5118   -0.2569    0.1769
               0.4579    0.2547    0.5728    0.1793    0.6043
               -0.1507   -0.5683   -0.1087    0.7682    0.2289

eigenvalues = -23.6752         0         0         0         0
               0      -3.6424         0         0         0
               0         0      0.0000         0         0
               0         0         0      41.9135         0
               0         0         0         0      629.4041
```

**c) Are their complex eigenvalues? (Check the same for more symmetric matrices). Explain why or why not?**

Check if there are complex eigenvalues.

```
if isreal(eigenvalues)
    disp('All eigenvalues are real.');
```

```
else
    disp('There are complex eigenvalues.');
```

```
end
```

**OUTPUT:**

All eigenvalues are real.

**d) How many eigenvalues of  $B$  are zero? Is there a relation between the number of zero eigenvalues of  $B$  and the  $rank(B)$ ?**

To count how many eigenvalues are zero.



```
zeroEigenvalues = sum(abs(diag(eigenvalues)) < 1e-6)
```

To check whether there is a relation between the number of zero eigenvalues of B

```
expected_Zero_Eigenvalues = size(B,1) - rankB
```

```
if zero_Eigenvalues == expectedZeroEigenvalues
    disp('The number of zero eigenvalues matches the expected relation with the rank of B.');
```

else

```
    disp('The number of zero eigenvalues does not match the expected relation with the rank of B.');
```

end

### **OUTPUT:**

zeroEigenvalues: 1

Expected\_Zero\_Eigenvalues = 1

The number of zero eigenvalues matches the expected relation with the rank of B.

### **e) Verify if the eigenvectors of $B$ are orthogonal.**

Check if eigenvectors are orthogonal.

```
orthogonalCheck = eigenvectors' * eigenvectors;
```

Check if eigenvectors' \* eigenvectors is close to identity matrix.

```
if norm(orthogonalCheck - eye(size(B,1))) < 1e-10
    disp('Eigenvectors are orthogonal.');
```

else

```
    disp('Eigenvectors are not orthogonal.');
```

end

### **OUTPUT:**

Eigenvectors are orthogonal.

### **6) Using MATLAB generate a $10 \times 5$ matrix $A$ of rank 3.**

**a) Obtain a symmetric matrix  $S1 = A * A^T$  and  $S2 = A^T * A$**

```
A = randi([0 9], 10, 3)*randi([0 9], 3,5)
S1 = A'*A
S2 = A*A'
```

**OUTPUT:**

```
S1 = 13357 11371 21381 21921 17363
      11371 10088 18023 20028 15034
      21381 18023 37595 36009 31227
      21921 20028 36009 41256 30978
      17363 15034 31227 30978 26544
```

S2 =

Columns 1 through 5

```
13461 14176 11618 12837 13466
14176 15068 11802 13706 14208
11618 11802 11475 10521 11828
12837 13706 10521 12500 12950
13466 14208 11828 12950 14325
11613 11770 11265 10408 10969
  438  488   264   446   353
20901 21808 18906 19717 21556
  9283 10006   7431  9196   9728
11266 11704 10088 10494 10842
```

Columns 6 through 10

```
11613 438 20901 9283 11266
11770 488 21808 10006 11704
11265 264 18906 7431 10088
10408 446 19717 9196 10494
10969 353 21556 9728 10842
11909 349 18251 6986 10512
  349  29   568   304   378
18251 568 33305 14399 17406
  6986 304 14399 6968 7318
10512 378 17406 7318 9800
```

**b) Find the rank of  $S1$  and  $S2$ .**

```
rank(S1)
```

```
rank(S2)
```

**OUTPUT:**

```
3
```

```
3
```

**c) Find the eigenvalues of  $S1$  and  $S2$ . What do you notice about them? Explain.**

```
EvalS1 = eig(S1)
```

```
EvalS2 = eig(S2)
```

**OUTPUT:**

```
EvalS1 = -0.0000
          0.0000
          0.0151
          0.0411
          1.2323
```

```
EvalS2 = -0.0000
          -0.0000
          -0.0000
```

0.0000  
0.0000  
0.0000  
0.0000  
0.0151  
0.0411  
1.2323

**OBSERVATION:** The symmetric matrices have same nonzero eigenvalues.

**7) Using MATLAB generate a  $7 \times 7$  matrix  $A$  of rank 2.**

**a) Obtain a skew-symmetric matrix  $B = A - A'$  and find rank of  $B$ .**

```
A = randi([0 9],7,2)*randi([0 9],2,7);
B = A - A'
rank(B)
```

**OUTPUT:**

```
B =  0  -42  31  22   7  -43  46
     42   0  55  49  70  -12  80
    -31  -55   0  15  14  -73   2
    -22  -49  -15   0  14  -66  -21
     -7  -70  -14  -14   0  -77  -21
     43  12  73  66  77   0  107
    -46  -80  -2  21  21 -107   0
```

Rank of B is 4.

**b) Find the eigenvalues and eigenvectors of  $B$  using MATLAB.**

```
[eigenvectors, eigenvalues] = eig(B)
```

**OUTPUT:**

```
eigenvectors =
0.2474 + 0.0092i  0.2474 - 0.0092i  -0.2292 + 0.5534i  -0.2292 - 0.5534i  0.0200 + 0.0000i  0.3976 + 0.0000i  -0.0360 + 0.0000i
0.3164 - 0.2815i  0.3164 + 0.2815i  0.2701 - 0.0994i  0.2701 + 0.0994i  -0.4818 + 0.0000i  0.1907 + 0.0000i  -0.4552 + 0.0000i
0.2287 + 0.2033i  0.2287 - 0.2033i  -0.1875 - 0.1028i  -0.1875 + 0.1028i  0.4128 + 0.0000i  -0.3060 + 0.0000i  -0.6762 + 0.0000i
0.1498 + 0.2324i  0.1498 - 0.2324i  0.1882 - 0.2462i  0.1882 + 0.2462i  0.3882 + 0.0000i  0.6976 + 0.0000i  0.1329 + 0.0000i
0.1647 + 0.2812i  0.1647 - 0.2812i  0.4885 + 0.2661i  0.4885 - 0.2661i  0.0479 + 0.0000i  -0.3857 + 0.0000i  0.1326 + 0.0000i
0.3593 - 0.3919i  0.3593 + 0.3919i  -0.0314 - 0.0726i  -0.0314 + 0.0726i  0.4295 + 0.0000i  -0.2465 + 0.0000i  0.4206 + 0.0000i
0.3283 + 0.3044i  0.3283 - 0.3044i  -0.2535 - 0.1913i  -0.2535 + 0.1913i  -0.5095 + 0.0000i  -0.1251 + 0.0000i  0.3495 + 0.0000i
```

```
eigenvalues =
0.0000 + 2.2664i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 - 2.2664i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.3597i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 - 0.3597i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
```

**c) Are their real eigenvalues? (Check the same for more skew-symmetric matrices). Explain why or why not?**

Check if eigenvalues are real (they should be imaginary for skew-symmetric matrices).

```

if all(isreal(diag(eigenvalues)))
    disp('All eigenvalues are real.')
else
    disp('Not all eigenvalues are real. There are imaginary eigenvalues.')
end

```

### OUTPUT:

Not all eigenvalues are real. There are imaginary eigenvalues.

### **d) Do you have one eigenvalue as zero? Can you explain why?**

Check if there's a one or zero eigenvalue.

```
has_zero_eigenvalue = any(abs(diag(eigenvalues)) < 1e-10)
```

### OUTPUT:

```
has_zero_eigenvalue = 1
```

### **e) In a similar manner generate a skew-symmetric matrix $C$ from an $8 \times 8$ matrix. Does $C$ have a zero eigenvalue? Why or why not?**

```
C = rand(8, 8) - rand(8, 8)'
```

Create an 8\*8 matrix and generate a skew symmetric matrix C.

Find the eigenvalues of C and check if it has a zero eigenvalue.

```

eigenvalues_C = eig(C)
has_zero_eigenvalue_C = any(abs(eigenvalues_C) < tolerance)
if has_zero_eigenvalue_C
    disp('C has no zero eigenvalue.')
else
    disp('C has a zero eigenvalue.')
end

```

### OUTPUT:

```

C =
-0.3608  -0.3757  -0.0333   0.0145   0.5572  -0.1614   0.1015   0.6048
-0.7036   0.0773   0.1045  -0.4991   0.0586  -0.5908   0.4842   0.1755
 0.3383  -0.0867   0.3766  -0.6664  -0.3513   0.2520   0.5264   0.2954
-0.4082   0.0096   0.4740   0.2680   0.3856  -0.3990   0.4382  -0.2506
 0.3451  -0.3535   0.0886   0.1388   0.3618   0.2702  -0.4332   0.4558
 0.7563   0.6153  -0.4182   0.3725  -0.1036   0.6320   0.5961   0.1862
-0.5805   0.4310  -0.8061   0.7127   0.6981  -0.4382   0.5997   0.2263
-0.4741  -0.5968   0.1185  -0.0270   0.4786   0.1469   0.2168   0.7574

```

```

eigenvalues_C =
1.2255 + 0.0000i

```

0.4577 + 0.9920i  
0.4577 - 0.9920i  
-0.0272 + 0.0000i  
0.0285 + 0.1454i  
0.0285 - 0.1454i  
0.2705 + 0.6001i  
0.2705 - 0.6001i

has\_zero\_eigenvalue\_C =  
0

C has a zero eigenvalue.

**8) Generate a matrix that will transform the points on a circular membrane to an elliptic membrane with a stretch factor of 5 and 9 along the directions  $[1,1]$  and  $[1,-1]$  respectively.**

Eigenvector matrix for directions  $[1,1]$  and  $[1,-1]$

$V = (1/\sqrt{2}) * \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix};$

Diagonal scaling matrix with stretch factors.

$D = \begin{bmatrix} 5 & 0 \\ 0 & 9 \end{bmatrix};$

Transformation matrix.

$T = V * D * \text{inv}(V);$

**OUTPUT:**

$T = \begin{bmatrix} 7 & -2 \\ -2 & 7 \end{bmatrix}$